Campus Santa Fe

Dimension Deck

Game Design Document

Course:

Software Construction and Decision Making

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Index

1. Game Design

1.1 Summary

A retro-styled roguelite adventure game with a 16-bit pixel art aesthetic, where every room of the map hides a surprise: a new card to collect, a hidden reward, or an unexpected boss fight. The player builds their deck as they explore, finding attack, defense, and support cards scattered across the thematic dimensions, creating a unique and evolving strategy with every run. Combat is real-time and entirely card-driven, meaning the hand you built determines whether you survive what's behind the next door.

Dimension Deck sits at the intersection of two genres without belonging fully to either. Unlike *Slay the Spire*, where combat is turn-based and purely strategic, and unlike *Neon White*, where cards are disposable movement tools, Dimension Deck makes cards and real-time movement equally necessary at all times. You cannot win by standing still and thinking, and you cannot win by running without a plan. The closest comparison is: *The Binding of Isaac* meets *Slay the Spire*, but where your deck actively fights back through automated synergies without you pressing a button.

1.2 Gameplay

Dimension Deck combines a fast-paced action roguelite with the strategic freedom of a deck builder presented in a 2D top-down view. It borrows the movement and spatial pressure of The Binding of Issac and layers on the car management depth of Slay the Spire.

The players use a deck of cards to materialize attacks, defenses, and effects in real time. The combat mechanics are split into two card categories: Active cards, which the player triggers manually to deal damage, heal or gain shields; and automatic cards, which fire on their own when defined trigger conditions are met, rewarding smart deck construction with passive power.

The player navigates procedurally generated rooms to find loot, shops, or enemies. The goal of each dimension is to clear enough rooms to reach and defeat the final boss, the transition into the next dimension.

Generation Rulebook

Each dimension is built from a sequence of rooms generated at the start of every run. A dimension always contains a minimum of 20 rooms and a maximum of 30 rooms before a Boss node is placed at the end. The exact number is chosen randomly within that range each time, so the player never knows how close they are to the boss until they reach it.

Rooms are not placed in a straight line.The generator builds a small bracing map where the player may have two or three possible paths to take at certain points, but all paths eventually converge at the Boss Node

Room Type Weighting

Room Type	Base Chance	Notes
Combat Room	High	Unlocks exists on full clear
Shop Room	Medium	Merchant with buy, sell and upgrade
Chest Room	Medium	Free random card reward or credits on interaction
Shrine Room	Low	Portal that transport the player to the next dimension
Glitch Room	Low	Contains a Glitch Pillar that grants a random card on interaction
Mr. Bombastic	Rare	A small Easter Egg

1.3 Mindset

We want the player to feel overwhelmed by the number of enemies on screen but empowered by the automated synergies their deck produces. The tension comes from managing movement and card rotation at the same time. A good run should feel like controlled chaos, where the player is always one smart card play away from turning a losing fight around. The emotional target is: *"I was about to die, but my deck saved me."*

2. Technical

2.1 Screens

Dimension Deck uses different core screens, each with its own identity and utility within the core gameplay loop. Screens are divided into three categories: Navigation and progression, screens, gameplay and customization screens and economy and result screens

Navigation and progression screens

Main Menu

- Purpose: Entry screen with player authentication (login / register via JWT), settings, credits and quit
- Elements: New Game, Continue, Settings/Options, Credits, and Quit.
- Visuals: Game logo, version number.

[IMAGEN del menu]

Dimension Map

- Purpose: A bird's-eye view of the current dimension showing the branching graph
- Elements: Nodes representing rooms types (Boss room, current room, shop room, undiscovered rooms, discovered rooms); connecting lines showing valid path

[IMAGEN del mapa generado]

Gameplay and customization screens

Room Screen

- Purpose: The main gameplay screen where all combat take place
- Elements: Enemy health bars, active card han at the bottom-left, cooldown wipe animations per card slot; player health, shield indicator and coins at the top-left

[IMAGEN completa del gameplay]

Deck Management Screen

- Purpose: Allows the player to review and organize their active and automatic card collections

- Elements: Grid view of all cards in the deck, action buttons for removing cards

[IMAGEN UI de cartas]

Economy and result screens

Shop Screen

- Purpose: Exchange credits for cards, sell unwanted cards and purchase slot upgrades
- Elements: Three tabs, Buy (5 slot with rarities distributions), Sell (lists all equipped cards with sell value), Upgrade (active slot and auto slot upgrade costs). Card carousel with rarity-colored glow on the center card

[IMAGEN UI de la tienda]

Game Over / Victory Screen

- Purpose: Summarizes the run and displays its outcome
- Elements: "Victory" or "Defeated" header, stats table with damage dealt, credits earned, and enemies slain, Return to Hub or Main Menu

2.2 Controls

Movement and Navigation

Key	Action
W / Arrow Up	Move the player upward
A / Arrow Left	Move the player to the left
S / Arrow Down	Move the player downward
D / Arrow Right	Move the player to the right
Mouse Cursor	AIM card effects toward the cursor position

Card Actions

Key	Action
Keys 1 - 3	Execute the corresponding active card slot (base hand)
Keys 4 - 5	Execute expanded active card slots
Left Click (LMB)	Confirm and execute the selected card at the cursor

Combat and Utility

Key	Action
Spacebar	Dash in the current movement direction
E	Interact with chest, glitch pillars, portals and shop
Tab	Toggle the deck management
Escape	Pause the game or close any open menu
Shift + H	Toggle debug hitbox visualization: yellow sprite bounds, red hitbox bounds, cyan center point
Shift + T	Toggle god mode: invincibility and zero card cooldowns

2.3 Cards

2.3.1 Card System

Cards are the foundation of both combat and progression in Dimension Deck. Every offensive and defensive action is executed through the player's deck, which is split into two distinct categories

Active Cards (Manual Execution): Active cards are tools that the player triggers manually, with numbers keys 1 - 5 or a left mouse click. They deal damage, restore health or provide defensive shields. Once activated, a card enters a cooldown period shown as a wipe overlay on the slot.

- Capacity: Starts at 3 slots, expandable to 5 via shop upgrades
- Subtypes: Melee (cone damage), Heal (restore HP), Defense (shield), Drain (damage nearest enemy, return HP)

Automatic Cards (Trigger-Based): Automatic cards sit in a separate pool and fire on their own when a defined trigger event fires. They have no cooldown and require no player input.

- Capacity: Supports between 4 and 8 slots.
- Supported triggers: On kill, on attack, on damage

Card Rarities

Rarity	Color	Description
Common	Grey	Base level cards found frequently
Rare	Blue	Improved cards with higher stats
Epic	Purple	Powerful mechanics that often serve
Legendary	Gold	Game-chanding finds that fundamentally

The Card Resistance System

Expanding the players' deck does two things: it increases the power, but also activates the Card Resistance system. As more card slots are unlocked, enemies gain damage, resistances, forcing the player to change their tactics and keep evolving through the run.

$$Resistance\% = (Activate\ Card\ Slots - 3) \cdot 5\%$$

Active Card Slots	Card Resistance
3 (base)	0%
4	5%

5 (max)	10%

The automatic card pool carries its own separate resistance. For every 2 automatic slots abose, the base 4, enemies gain an additional 3% resistance

Active Card Slots	Card Resistance
4 (base)	0%
6	+3%
8 (max)	+6%

Both values stack. A player with 5 active slots and 8 automatic slots faces a total of 10% + 6% = 16% damage reduction on all enemies in the current room.

2.3.2 Card Acquisition

Starter Deck: Every new run begins with three cards: Quick Strike (active melee, common), Heal Pulse (active heal, common) and Wood Shield (active defense, common)

Field Rewards: Chest Room contains a random card reward. Glitch Room contain a Glitch Pillar that grants a random card on contact, the card goes directly into the deck with no selection screen

The Dimension Store (Shop): The primary hub for deck management

- Credits: Credits (CR) dropped by enemies and chests
- Stock: 5 slots per visit, generated via SLOT_RARITIES with distribution: [common/rare], [rare], [rare/epic], [epic/legendary], [any].
- Sell values: Common 30 CR, Rare 70 CR, Epic 130 CR, Legendary 220 CR
- Slot upgrades: Active slot upgrades costs 150 CR (3 to 4 slots) and 280 CR (4 to 5 slots). Auto slot upgrades costs 110 / 160 / 210 / 270 CR per tier

2.3.3 Card Upgrades

Every card has three levels: base (1), upgraded (2), and max (3). Cards start at base when obtained. Acquiring a second copy of the same card triggers an automatic merge that advances it to level 2. A third copy advances it to level 3 (max).

When a max-level card receives a duplicate, the extra copy converts into credits instead of merging. Credit value by rarity: Common = 10 CR, Rare = 25 CR, Epic = 50 CR, Legendary = 100 CR.

Active Cards:

- Level 2: Reduces the card's cooldown by 30%
- Level 3: Increases base damage, healing or defense value by 40%

Automatic Cards:

- Level 2: Widens the trigger condition to activate in more situations
- Level 3: Increases effect potency, larger damage, longer buff duration

2.3.5 Combat: Card Rules

Active Card Execution: Pressing the corresponding number key triggers the card's effect immediately. There is no cast time or mana cost, allowing seamless integration with movement. Once activated that slots dims and circular animation shows remaining recharge time

- Common cards: short cooldown (2 - 6s) for consistent, repeatable use
- Legendary cards: long cooldowns (10 - 25s) saved for boss phases or being overwhelmed

Damage Resolution: Melee cards deal damage to all enemies whose hitbox falls within a cone-shaped arc in the players' aim direction. Drain cards target the nearest living enemy.

2.3.5 Card Catalog

Active Cards – Starters

Name	Rarity	Cooldown	Effect
Quick Strike	Common	2s	Deal 25 damage to enemies within 80px in a forward cone
Heal Pulse	Common	5s	Restore 25 HP
Wood Shield	Common	6s	Absorb the next 20 damage

Active Cards

Name	Rarity	Cooldown	Cost	Effect
Iron Fist	Rare	5s	120 CR	55 Damage within 48px range
Blood Siphon	Rare	10s	140 CR	45 Damage to nearest enemy, restore 20 HP
Stone Wall	Rare	10s	125 CR	Absorb the next 50 damage
Remedy Vial	Rare	8s	130 CR	Restore 42 HP
Nova Burst	Epic	9s	210 CR	110 damage within 72 px range
Mending Wave	Epic	12s	220 CR	Restore 85 HP
Mirror Guard	Epic	14s	200 CR	Gain 58 shield
Shadow Blade	Legendary	10s	360 CR	250 Damage

Phoenix Elixir	Legendary	18s	380 CR	Full restore al HP
Diamond Fortress	Legendary	15s	350 CR	Absorb the next 100 damage

2.3.7 Card Design

2.4 Combat

2.4.1 Combat Overview: Real-Time Tactical Movement

Combat in Dimension Deck is defined as a real-time, continuous experience where enemies spawn and attack immediately upon the player entering a room. The system requires simultaneous attention to spatial awareness and card-based decision-making.

- Movement and Defense: Players use WASD or arrow keys for top-down directional control to physically dodge projectiles and melee strikes.
- The Dodge Mechanic: The spacebar triggers a dash, providing a reliable escape tool when the player is surrounded.
- Card Execution: Players aim with the mouse and execute active cards using number keys 1 through 5.
- Telegraphed Danger: Enemy attacks are telegraphed via wind-up animations or visible paths projected on the floor, giving the player seconds to react.

2.4.2 Enemy Behavioral Archetypes

All enemies are built upon three foundational archetypes that dictate their AI and physical presence:

- Swarm Archetype
  - Characteristics: These are small enemies, typically rendered at 16 x 16 px.
  - Behavior: They use a "seek-and-close" movement pattern to pursue the player.
  - Combat Role: They pressure the player by crowding the playable space, forcing constant movement to avoid contact damage.
- Tank Archetype
  - Characteristics: Large, high-health sprites that communicate a high level of threat.
  - Behavior: They move slowly but are highly resistant to knockback.
  - Combat Role: They act as obstacles that deal massive damage on contact, requiring the player to maintain distance.
- Ranged Archetype
  - Characteristics: Mid-sized enemies that prioritize staying away from the player.
  - Behavior: They maintain a safe distance and track the player to fire projectiles on a cooldown.
  - Combat Role: They force the player to dodge across the room while trying to find an opening to counter-attack.

2.4.3 Dimension-Specific Bestiary

The Dark Ages

Enemy	**Archetype**	**Combat Stats & Behavior**
Dungeon Rat	Swarm	Fast-moving; relies on numbers to corner the player.
Slime	Swarm	Slows player movement or leaves hazards on the stone floor.
Skeleton	Tank	High HP; uses slow but heavy, telegraphed melee swings.

Old Western

Enemy	Archetype	Combat Stats & Behavior
Desert Rat	Swarm	Aggressive small pests found near dry bushes and cacti.
Bandit	Ranged	Strafes across the sand while firing pistol shots.
Cactus Thug	Tank	High defense; punishes players who get too close with needle-based contact damage.

2.4.4 Strategic Progression: Card Resistance

To ensure the game remains challenging, the Card Resistance system scales with the player's power. As the player expands their card slots or builds a larger deck, enemies gain resistance to specific card types, preventing a single strategy from dominating the entire run.

2.5 Roguelite Progression and Telemetry

Runs are independent, and permanent death resets the player back to the Hub World with their current deck lost, requiring a fresh start for the next attempt. However, dimensional credits earned during a run are carried over and can be spent on permanent upgrades that expand stats and maximum card capacity across future runs. These Hub upgrades ensure long-term empowerment without trivializing individual runs.

The game collects telemetry data across four areas to support ongoing balance work. Player performance tracking monitors run duration and the highest biome reached to identify difficulty spikes. Win/loss analytics record the frequency of Game Over versus Victory screens to gauge overall accessibility. Card and trigger usage analyzes which active cards are chosen most often and which automatic triggers prove most effective in combat. Economic balancing logs dimensional credit earnings to see whether players prioritize new cards, shop items, or permanent Hub upgrades. All collected data feeds back into dynamic scaling adjustments to Card Resistance, ensuring enemies stay competitive as the player's deck capacity grows.

Three specific KPIs the game will track:

1. **Time per dimension**: Average minutes spent in Old West vs. Dark Ages per run. This identifies which biome causes players to stall or rush, pointing to pacing issues.
2. **Most discarded card**: Which active card gets removed from the deck most often at the shop. A card discarded frequently signals it underperforms relative to its cost.
3. **Run length at death**: The room number where most Game Over screens occur. This pinpoints difficulty spikes in the procedural layout.

3. Level Design

3.1 Themes

3.1.1 Ambience

Dimension Deck uses a 16-bit visual style. This design reminds players of classic gaming eras while providing a fluid and nostalgic experience. The game features two distinct dimensions, each with its own look.

- **The Old West:** This theme uses dusty deserts and frontier towns to create a sense of lawlessness and freedom. The wide-open space of the desert provides a feeling of freedom that stands in total contrast to the next world.
- **The Dark Ages:** This world is inspired by fantasy games like Dungeons and Dragons and takes place inside a castle and caverns. This setting makes the player feel trapped in a heavy, enclosed environment.

Using these two themes helps prevent the player from getting bored with the visuals. It also allows for unique creative additions in each biome without breaking the player's feeling of being in the game or the core gameplay experience.

3.1.2 Objects

3.1.2.1 Ambient

Player Character: The player uses a **32 x 32 px** sprite. This larger size makes the hero stand out clearly against the background and enemies.

Theme Elements: Decorative items like desert plants for the West or stone pillars for the Dark Ages help ground the player in each dimension.

3.1.2.2 Interactive

Each interactive object follows defined rules for how the player engages with it:

- **Card Reward Chests**: Appear after a room is fully cleared. Require no key. The player presses E to open and receives a choice of 3 cards drawn from the current biome's rarity pool. If the player skips the chest, it disappears when they exit the room.
- **The Dimension Store**: A safe zone; no enemies spawn here. The shopkeeper offers 4 cards for sale per visit, randomly pulled from common and uncommon pools. Prices range from 20 to 80 dimensional credits depending on rarity.
- **Glitch Pillars**: The player touches them to interact; no button press needed. Each pillar either reveals a hidden exit or upgrades one common card in the player's current hand to uncommon. Only one pillar upgrade is allowed per run.
- **The Card Smith's Anvil**: Requires two copies of the same active card. After merging, the card's cooldown is cut by 40%. The original two copies are consumed.
- **Dimensional Altars**: Removes one automatic slot permanently for that run. In exchange, all active card damage increases by 30% for the remainder of the run. This choice cannot be reversed.
- **Dimension Shift effect on objects**: When the player crosses into a new dimension, all chests and pillars from the previous dimension are removed. Credits and deck contents carry over; room objects do not.

3.1.3 Challenges

Dimension Deck's challenges aim to evaluate players' reflexes and their capacity to devise a card strategy amidst relentless pressure.

- Real-Time Combat: During each run the enemies move and attack the moment they see the player. With this mechanic, we make the player stay in constant alert to dodge or protect incoming projectiles and melee hits while also trying to play their cards in order to deal damage to the enemies.
- Attack visualization: Most of the enemies will have animations or visible paths on the floor, giving the player seconds for them to react and dodge the incoming attacks.
- Card Resistance: As the deck gets bigger and the player gets stronger, enemies will start to develop resistances to certain types of cards that the player has during that run.

This mechanic stops the player from using the same strategy the whole game, forcing him to change the idea the player is focusing on for completing each level.

- Boss Battles: At the end of each level, the player must fight a boss that changes its behavior and attacks the player as the fight goes on. These fights test how well the player has built their deck and how good their skills at the game are.
- Procedural Rooms: Each room will be randomized, making the player never know if the next room will be full of dangerous enemies, traps, loot or the level boss to finish the level and continue the run.
- Permanent Death: If you run out of health, the run ends and you lose your current deck, sending you back to the start to try again with a fresh strategy.

3.2 Game Flow

The flow of Dimension Deck focuses on moving through a series of unpredictable rooms, where the main goal of the player is to survive and find an exit to the next challenge. Here is how the game progression works:

- Start of the run: A run begins in the Dark Ages or the Old West, making each run feel unique from any other. The player enters the first room with a starting deck, full directional control and free decision-making.
- Clearing the rooms: In order to move forward, players must clear the current level they are on, which can be done in different ways. This could be achieved either by eliminating the room's boss or finding a rift that acts as a gateway to the next area. Once the level is cleared, the room could drop rewards like the dimensional credits.
- The final boss encounter: The final boss of a dimension does not appear at a fixed level; it can be encountered randomly, making the player never know when they will face the most difficult enemy of the run. The difficulty of this encounter scales dynamically using the Card Resistance system, meaning the boss becomes tougher if the player has built a larger, more powerful deck.
- Post-Level Store: After successfully finishing a level, the player enters the Dimension Store. This is a safe area between levels where players can spend credits on new cards, buy more slots for their deck, and manage their primary and secondary card hands.
- Deck Management and Storage: Players are able to see and manage their deck at any time, but the store is the best place to reorganize it. If a player has more cards that their current slot allows, they can store them in their collection, though carrying a larger deck increases the resistance enemies have against their attacks.

- Dimension Transition: Once the final boss is defeated, a portal opens to transport the player into the next dimension. This transition moves the player into a different environment with new visuals, enemies, and cards.
- The Death/Victory Cycle: If the player loses all health, the run ends and the current deck is lost. However, any credits earned are carried back to the Hub World to purchase permanent stat and capacity upgrades, making the character stronger for the next attempt.

4. Development

The codebase is organized around a set of abstract base classes that define shared behavior without implementing it, keeping individual systems extensible and the overall architecture clean.

- Entity: The root class for anything that exists in the game world. Holds position, velocity, health, and a basic update/render cycle. All moving, interactable objects extend this.
- Card: Defines the structure every card must follow: a name, a rarity, an energy cost, an effect function, and a card type flag (active or automatic). The effect function takes a reference to the current combat state and resolves the card's outcome. Cards also carry a cooldown value used by the discard pile timer system.
- Enemy: Extends Entity and adds abstract methods for movement behavior, attack patterns, a Card Resistance value that scales with the player's deck capacity, and a drop table reference. Every enemy type implements these differently.
- Room: Defines the container structure for a playable space: fixed dimensions of 480 x 352 pixels across a 15 x 11 tile grid, spawn points, exit conditions, and lists of enemies and objects. Room subtypes inherit from this and override spawn logic and exit triggers.
- Biome: Groups a set of room types, an enemy pool, a visual theme identifier, and a boss reference into a single traversable dimension. Controls which rooms can appear on the Dimension Map for that biome and manages the Card Resistance multiplier applied to all enemies within it.
- Relic: Defines passive items with a trigger condition and an effect callback. The base class handles registration with the combat event bus; derived classes define what triggers them and what they do.

- Synergy Effect: Defines the structure for combination effects between cards. Holds a trigger card reference, a catalyst card reference, a multiplier value, and a result effect function. Registered with the Synergy Manager at runtime.

4.1 Derived Classes

Derived classes implement the concrete behavior defined by each abstract parent. Below are the primary derived types used across the game's systems.

- Player (extends Entity): The player character is rendered at 32 x 32 pixels and changes visual design depending on the active dimension. Adds the active card slot manager (3 slots, expandable to 5), the automatic card pool (4 to 8 slots), hand management, dimensional credit tracking, and the dodge mechanic. Handles collision with pickups and interaction prompts for objects. Listens to the input manager and translates inputs into movement vectors and card plays.
- StrikeCard, AreaCard, HealCard (extend Card, active type): The three foundational active card types. StrikeCard resolves direct damage against a single target. AreaCard applies damage across a radius centered on the cursor position. HealCard restores a portion of the player's missing health and enters a longer cooldown than offensive cards to prevent it from replacing resource management.
- ConditionCard (extends Card, active type): An active card that applies a status effect to one or more enemies and queues a synergy flag in the SynergyManager. The Oil card is an example: it applies a flammable condition that the SynergyManager monitors, ready to trigger a tripled damage multiplier when a Fire active card resolves on the same target.
- TriggerCard (extends Card, automatic type):Sits in the automatic pool and watches the combat event bus for its trigger condition. When met, it fires its effect function without player input. Examples include granting a combat buff when entering a boss room, or creating a defensive explosion when the player takes critical damage.
- BoostCard (extends Card, automatic type): A permanent stat modifier for the duration of the run. Adds its value to a stat register when added to the deck and subtracts it on removal. Has no event trigger and no cooldown it contributes passively at all times.
- SwarmEnemy, TankEnemy, RangedEnemy (extend Enemy): The three base enemy behavioral archetypes. Enemy sprite dimensions vary by type: small swarm enemies render at 16 x 16 pixels and use a seek-and-close movement pattern; tank enemies are larger, move slowly, resist knockback, and hit for high damage on contact; ranged

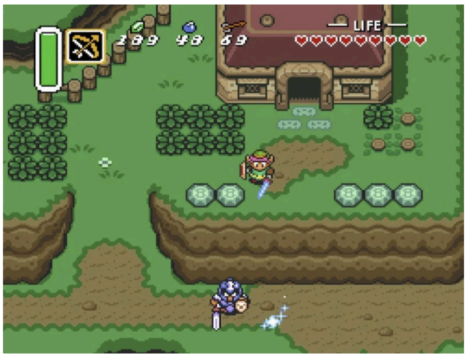
enemies maintain distance, track the player, and fire projectiles on a cooldown. All three archetypes change visual design depending on the active dimension.

- BossEnemy (extends Enemy): Adds a phase list, a phase transition trigger, and a secondary attack pattern that activates in later phases. Also holds a unique mechanic flag that activates a special rule for the boss fight. Boss sprites scale up significantly from standard enemy sizes to communicate their threat on screen.
- CombatRoom, ShopRoom, ChestRoom, ShrineRoom (extend Room): Each overrides the spawn logic and exit condition appropriate to its type within the fixed 480 x 352 pixel, 15 x 11 tile grid. CombatRoom spawns enemies from the biome pool and unlocks exits on full clear. ShopRoom spawns the merchant object and allows free exit after browsing. ChestRoom spawns one chest tied to a rarity roll and exits after interaction. ShrineRoom spawns one shrine and exits after the player accepts or declines the offer.
- SynergyManager: Singleton class that registers all SynergyEffect instances at startup and listens to card resolution events during combat. When a trigger card resolves and sets a synergy flag on a target, the SynergyManager holds the flag for a defined window. If a matching catalyst card resolves on the same target within that window, the multiplier is applied and the flag is cleared.
- TelemetryLoggerPassive singleton that hooks into key game events without affecting gameplay logic. Logs run duration, biome reached on death, Game Over and Victory frequencies, card selection rates, automatic trigger activation rates, and dimensional credit spending patterns. Data is written to a local session file and can be exported for balance review.

5. Graphics

5.1 Style Attributes

The game has a 16-bit aesthetic, using references from Super Nintendo Games such as The Legend of Zelda: A Link to the Past, The Escapists, Mario & Luigi Sagas or Pokemon. Each region of the map has a unique characteristic which gives a visual feedback to the player to know where is him. Style graphic is pixel art in 2D, with resolution of 16x16 bits for each tile.



Illustrative Image from ToZ: A Link to the Past

Cards Display (bottom center): The six cards available during the game will be displayed in the lower central area. These cards are pre-selected by the player. Cards can be selected using the mouse with a left-click. Selected cards will be highlighted with a blue border and the player can choose to use multiple cards at once

Player health (bottom right) : Displays the current health of the player

Deck menu (bottom left): This is the main menu of the game. When opened, it shows the player's deck, allowing them to select the six cards they will use in-game. The player can also enable or disable cards that function as passive abilities

Illustrative Image: Player uses a card against an enemy

Visual feedback will be used for all important actions occurring in the game. When the player takes damage, the screen will display a brief red flash. As the player's health decreases, the intensity of this flash will increase. Enemies, when taking damage, will briefly change their color to red for less than half a second

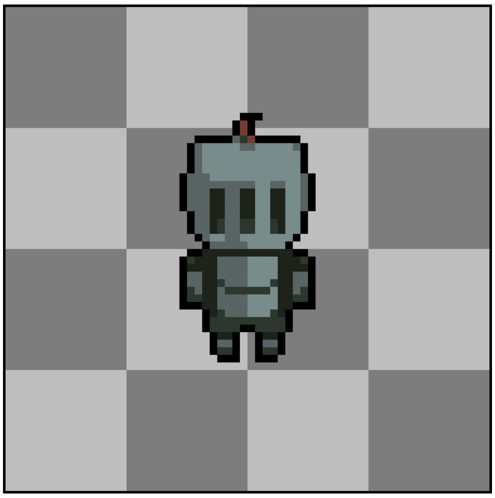
When a card is obtained, an image of the card will be displayed, drawing the player's attention as the main focal point

5.3 Sprites

Player:

- Dimension: 32 x 32 px
- Idle in 4 directions: left, right, up, down
- Walking in 4 directions: left, right, up, down

- Take damage
- Death



Enemies: Default movements

- Dimension: Depends on the enemy type
- Idle in 2 directions: left, right
- Walking in 2 directions: left, right
- Attacking in 2 directions: left, right

Tiles:

Each dimension will have a base map with a resolution of 480 x 352 px. From this, a tileset will be used by the system to procedurally determine the placement and distribution of tile objects and decorations across the map

1. **Old West**
   a. **Terrain**
      i. Sand

ii. Dirt path

iii. Rocky ground

iv. Small elevation

b. Natural props

i. Cacti

ii. Dry bush

iii. Animal bones

c. Interactive Objects

i. Boxes

ii. Chests

d. Obstacles

i. Rocks

2. Dark Ages

a. Terrain

i. Stone floor

ii. Pillars

iii. Doors

b. Natural props

i. Torches

ii. Tables

iii. Blood

iv. Bones

c. Interactive objects

i. Chests


6. Sounds/Music

6.1 Style Attributes

The soundtrack is a core component of the game experience. It is built around a 16-bit aesthetic, using synthesized sound and short looping compositions that reinforce the repetitive style of a roguelite game

The primary goal of the soundtrack is to generate a constant sense of tension, communicating to the player that they are in a dangerous environment where enemies can appear at any moment and potentially end the game. The music does not evolve with the player progression, instead, it maintains sustained pressure. Each dimension has it own distinct soundtrack identifies to differentiate environments and reinforce immersion

Old West: Inspired by classic westerns and desert styles. The music conveys isolation, vast open spaces and latent danger. It uses slower tempos and conveys a sense of isolation, vastness and latent danger:
https://www.youtube.com/watch?v=KhChHmk1o8c

Dark Age: Inspired by dark fantasy dungeons settings and classic games such as The Legend of Zelda, conveying a sense of confinement and constant threat:
https://www.youtube.com/watch?v=Nxx3Ti83TYk

6.2 Sounds Needed

All sound effects follow a retro 16-bit aesthetic to maintain consistency with the visual and musical identity of the game. Sound effects are grouped based on their function:

Interaction sounds:

- Chest opening: https://www.freesound.org/people/jon1/sounds/95503/
- Impacts against walls or objects: https://www.freesound.org/people/toxicwafflezz/sounds/150838/

Player and enemy feedback:

- Damage effect (player): https://www.freesound.org/people/mrickey13/sounds/515624/
- Damage effect (enemy): https://www.freesound.org/people/Mrthenoronha/sounds/506586/

Each sound must clearly differentiate between player and enemy actions

UI Sounds:

- Card reveal
- Card played
- Menu selection https://www.freesound.org/people/ZeltBolt/sounds/833055/

Ambient Sounds:

- Desert wind https://www.freesound.org/people/felix.blume/sounds/156414/
- Whispering distant voices

Ambient sounds help reinforce the atmosphere of each dimension and should remain subtle to avoid overwhelming the player

6.3 Music Needed

The game requires different music tracks depending on the gameplay state

Core Gameplay

- Dungeon exploration https://www.newgrounds.com/audio/listen/625484
- Combat https://www.newgrounds.com/audio/listen/76637

Key Event

- Final boss battle https://www.newgrounds.com/audio/listen/436762

UI / Game States

- Main menu https://www.playonloop.com/2020-music-loops/misty-dungeon/
- Defeat screen https://www.newgrounds.com/audio/listen/368887
- Victory screen https://www.newgrounds.com/audio/listen/1323003
- Credits screen https://www.newgrounds.com/audio/listen/11684

Special Content

- Special room

7. Schedule

The game development is organized into 8 stages and will be carried out in a staggered and progressive manner, maintaining a modular development that allows for system scalability with the aim of facilitating easier system debugging. Each system should only do one job and not several. The team consists of 3 developers, and collaborative work will be done with GitHub. The presented timeline maintains a clear and concise structure, prioritizing the individual development of each element for its integration at the end of each stage. The team seeks to perform continuous tests and adjustments for the early detection of problems that could hinder or compromise the complete development of the system.

Week 1 – Core System

- Game object class development
- Implement of:
    - Player class (basic movements and initial attributes)
    - Enemies class
- Input system implementation
- Basic physics and collisions system
- Initial structure card system
- First version of the game loop

Goal: Establish foundation game and minimal playable game

Week 2 – Gameplay and Cards system

- Completed card system development
    - Active card system
    - Passive card system
- Implementation of card mechanics (damage, effects)
- Enemy development (basic movement and logic)

Goal: Implement the main gameplay mechanics

Week 3 – Procedural Map Design

- Development of the basic procedural map generator
- Implementation of:
    - Enemy AI
    - Store system (buying card)
    - Economic system (currency)
- Progression system and difficulty scaling
- Design of cards and enemies

Goal: Procedural map generator, game logic and dynamic content

Week 4 – UI / UX

- UI development:
  - Game menu (main menu, pause menu)
  - Inventory system
  - Victory screen and Death screen
  - Deck UI (card management interface)
- UX feedback
  - Player visual indicator (damage, effects, interactions)
  - Player action feedback
- Integration of sound effects (combat, background sounds, actions)
- Refinement game loop
- Gameplay balancing
  - Gameplay flow
  - Player feelings
  - Cards balancing
  - Economy

Goal: Improve user experience and visual final finishes

Week 5 – Testing, debugging and final delivery

- Gameplay testing
  - Internal testing
  - External player testing
- Bug fixing
- Development of website
- Final build and presentation

Agregar

- Bitácora
- Estadísticas
- Efectos de cartas
- Last update en cada tabla
- Agregar informacion en torno a user